LESSON 01 · MISSION GUIDE

Priya's Inbox

Numbers, text and formatting: the bread and butter of every payroll.

From **Priya, Payroll**
To **Jen**
Subject **My inbox (help)**

Morning Jen! Hope the keycard works. Quick update from Payroll:

MACRO-9000 crashed over the weekend with "Run-time error '13': Type mismatch", which is Dave's way of saying hello. I've got seven tickets that need doing: pay maths, a messy staff export, payslips Gerald thinks look "a bit 1997", and numbers that the HR system insists are text. Oh, and Gerald wants the gender pay gap "in one sentence" for the board on Thursday.

Mo says Python can do all of this. Mo also said the printer would be fixed by Tuesday. No pressure! — P.

What you'll be able to do after today

- [ ] Store values in **variables** and calculate with them: `+ - * /`, plus `//`, `%` and `**`
- [ ] Clean up messy text: trim spaces, fix capitals, replace characters
- [ ] Turn text like `"£32,500.00"` into a number you can calculate with, and back again
- [ ] Format output like Excel number formats: `£2,708.33`, `12.5%`, neat columns
- [ ] Ask yes/no questions with `True` and `False`
- [ ] Cut text apart like `LEFT`, `RIGHT` and `MID`, but counting from 0

Why this matters at work

- **Every export arrives as text.** CSVs, HR systems, web forms: all text, even the salaries. Converting it properly is step one of every analysis.
- **Payslips and reports are formatted numbers.** If you can make one payslip line look right, you can make 4,000 of them look right.
- **The pay gap is just arithmetic.** Exercise 7 is the same formula the final project uses on the whole payroll.

Four big ideas

1. No `Dim`	`salary = 32500` creates the variable. Python works out the type itself.
2. Everything has a type	`int` (42), `float` (37.5), `str` ("Jane"), `bool` (True). `type(x)` tells you which.
3. Python never guesses	Excel turns "5" into 5 behind your back. Python refuses: `"5" + 3` is an error. You convert on purpose with `int()`, `float()`, `str()`.
4. Counting starts at 0	The first letter of `"Jane"` is `"Jane"[0]`. Excel would say position 1.

Three VBA habits that will bite you

- `^` is **not** "to the power of". Use `**`. (`2 ^ 10` is 8. Don't ask.)
- `=` stores a value, `==` compares. VBA uses `=` for both.
- Names are **case-sensitive**: `Salary` and `salary` are two different variables.

How today works

15 MIN
Lecture
Lars demos everything live, mistakes included

60 MIN
Exercise sheet
Seven tickets from the inbox. You write every file yourself

15 MIN
Dave's Leftovers #1
Three bugs. HMRC is waiting

The inbox

#	FROM	TICKET	YOU'LL USE
1	Priya	Pay maths	`/ // % ** round()`
2	Mo	Clean up the staff export	`.strip() .title() .lower() .replace()`
3	Gerald	A payslip that isn't "a bit 1997"	f-strings: `f"£{x:,.2f}"`
4	Priya	Text that should be numbers	`float() int() type()`
5	Siobhan	Yes or no	`== <= and or not`
6	Mo	LEFT, RIGHT, MID	`text[:3] text[-4:] .find()`
7 ⭐	Gerald	The pay gap in one sentence	all of the above

Your three documents

GUIDE.pdf	This one.
EXERCISES.pdf	The exercise sheet: what to write, and the expected output to check against.
CHEATSHEET.pdf	Everything from today on one page. Keep it open while you work.

You're done when...

- [ ] `my_work/` has `ex1` to `ex6` (and `ex7` if you're feeling brave), each printing the expected output
- [ ] Dave's script prints the expected two lines, with one commit per bug
- [ ] Everything is pushed to `lesson-01` and you've opened a pull request



Next lesson: The Bonus Run. Siobhan has 200 employees and a set of bonus rules that Gerald changes every Friday. You'll teach Python to make decisions (`if`) and to repeat things without copy-paste (`for`).